Data Sheet

EXPERIMENT NO. 03

Title: - Simulation of Transistor Circuit with Fixed-Bias Configuration and Study of Output Parameters (using Tinkercad/Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus Component	Specification	Quantity
1	Resistor	1KΩ, 2KΩ, 3KΩ	2
2	Capacitor	100 mf	2
3	Battery	15v	1
4	Ground	–	1
5	Transistor		1
6	Oscilloscope	50ms, 1v	1
7	Function Generator	10Hz, 6V, Sine	1
8	Connecting Wires	–	6
9	Breadboard	–	

Important Formulae: -

Base Current (I_B): -

$I_B = (V_{CC} - V_{BE}) / R_B$

- V_{CC} = Supply voltage
- V_{BE} = Base-emitter voltage (0.7 V for Si transistor, 0.3 V for Ge transistor)
- R_B = Base resistor

Collector Current (I_C): -

$I_C = \beta \times I_B$

- β = Current gain (transistor parameter)

Collector-Emitter Voltage (V_{CE}) — Theoretical: -

$V_{CE} = V_{CC} - I_C \times R_C$

Stability Factor: -

Expt.No. 3

Page No.
Date.

Title :- Simulation of Transistor Circuit With fixed - Bias Configuration and Study of Output Parameters (using TinkerCad / Proteus Software)

Theory :- The fixed-bias Configuration Set the DC operating point (Q point) of a BJT by Connecting the base to V_{cc} through R_B, the Collector through R_c and grounding the emitter.

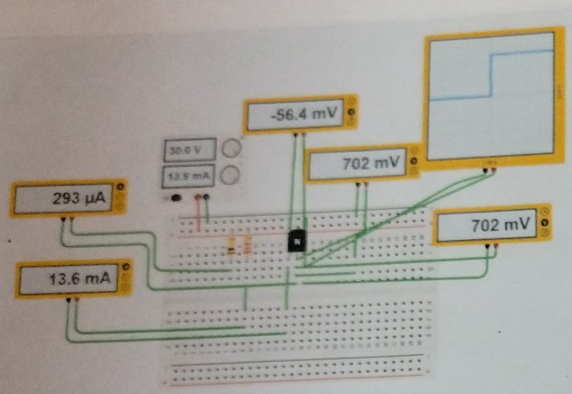
- Base Current: $I_B = (V_{cc} - V_{BE}) / R_B$ ($V_{BE} \approx 0.7$ V)
- Collector Current: $I_c = \beta \cdot I_B$
- Collector - Emitter Voltage: $V_{CE} = V_{cc} - I_c R_c$
- Stability factor: $S = \beta + 1$

Instrument & Component Required :-

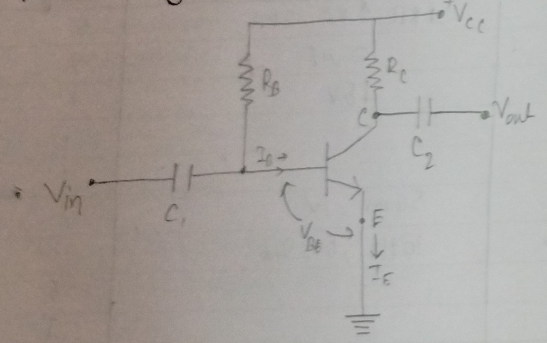
Sl.No	Components	Specification	Quantity
1	Resistor	1KΩ, 2KΩ, 3KΩ	2
2	Capacitor	100 MF	2
3	Battery	15V	1
4	Ground	–	1
5	Transistor	–	1
6	Oscilloscope	50ms, 1v	1
7	Function Generator	10Hz, 6V, Sine	1
8	Connecting Wires	–	6
9	Bread Board	–	1

OXFORD

Teacher's Signature



Circuit Diagram :-



Expt.No.

Page No.
Date.

Observation Table :-

Sl.NO	I_B (uA)	I_C (mA)	R_e (Ω)	V_{CE} (Practical) (V)	V_{CE} (Theoritical) (V)
1.	23.83	2.38	1 KΩ	12.55	12.62
2.	23.83	2.38	2 KΩ	10.10	10.23
3.	23.83	2.38	3KΩ	7.65	7.85

Calculation :-

$I_B = 23.83$,

$I_C = \beta \times I_B =$ 2.38

V_{CE} (Theoretical) $= V_{cc} - I_e \times R_e = 10.23$

% Error between Practical and Theoretical (V_{CE}) $= [(V_{CE(Th)} - V_{CE(Pr)}) / V_{CE(Th)}] \times 100$
$= 1.27\%$

$I_B = \frac{(V_{cc} - V_{BE})}{R_B} = \frac{15 - 0.7}{6.8 \times 10^5} = 23.83\ mA$

$I_e = \beta \times I_B = 100 \times 23 = 2.38\ mA$

$V_{CE} = V_{cc} - I_e \times R_e = 15 - (2.38 \times 2) = 10.23V$

% Error $= \frac{[10.23 - 10.10]}{10.23} \times 100\% = 1.27\%$

Conclusion :-

The fixed bias transistor circuit was Successfully Stimulated. The practical values closly match the thoritical calculations, yielding a minimal error of 1.27% for $R_e = 2K\Omega$ @V. This validates the theoritical DC bias equations and Confirms the increasing R_e reduces V_{CE}, driving the transistor closer to Saturation.

OXFORD

Teacher's Signature